

Skills Development for Adolescent Girls

Investment case, East Java, Indonesia

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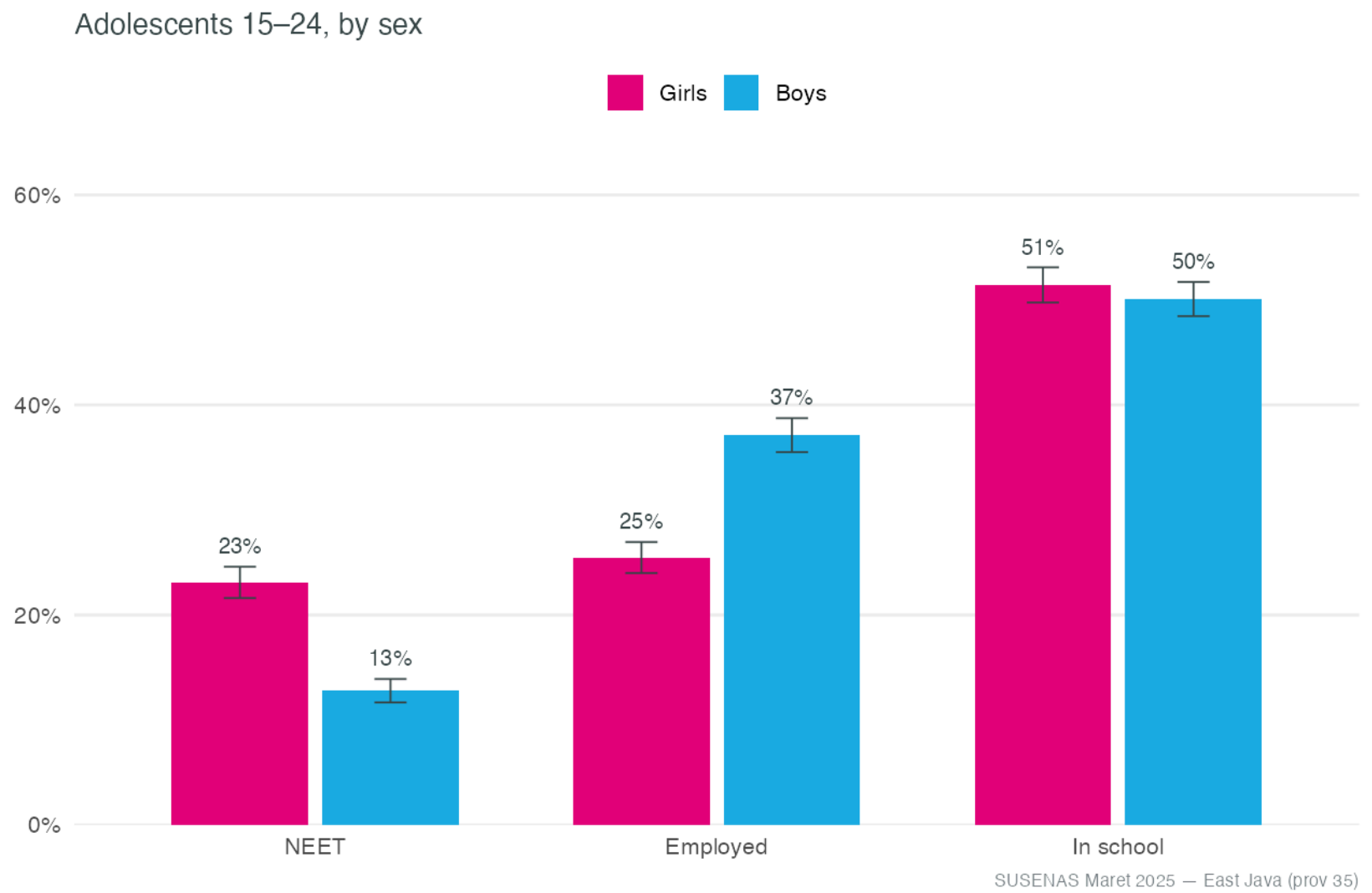
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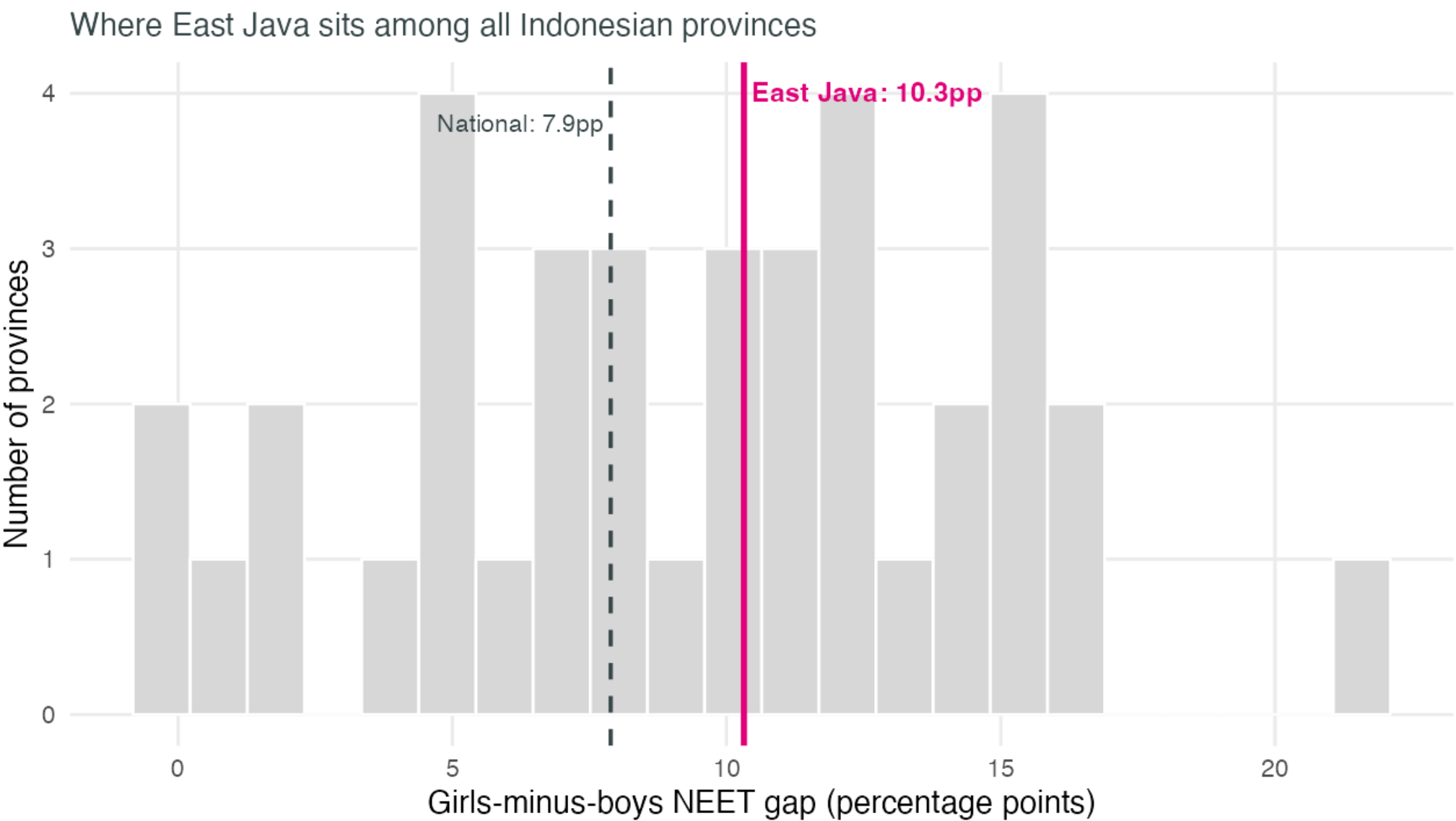
1 • The problem

Girls attend school at the same rate as boys, but their NEET rate is close to double



Girls 15–24: 51% in school (boys 50%), 23% NEET (boys 13%), 25% employed (boys 37%). The gap opens at the transition out of school. SUSENAS Maret 2025, survey-weighted, 95% CIs.

East Java's NEET gender gap is wider than the national average



SUSENAS Maret 2025, all provinces, survey-weighted; adolescents 15-24

Girls-minus-boys NEET gap: 10.3 points in East Java against 7.9 points nationally. Same survey, same definition, all provinces.

The gap costs an estimated USD 4.1 billion in forgone lifetime productivity

Forgone lifetime productivity from the NEET gender gap

Excess NEET girls (10.3pp gap above the boys' rate) x present value of lifetime earnings

Per excess-NEET girl (USD)

USD 14,143



Province-wide, today (USD millions)

USD 4,077.0M

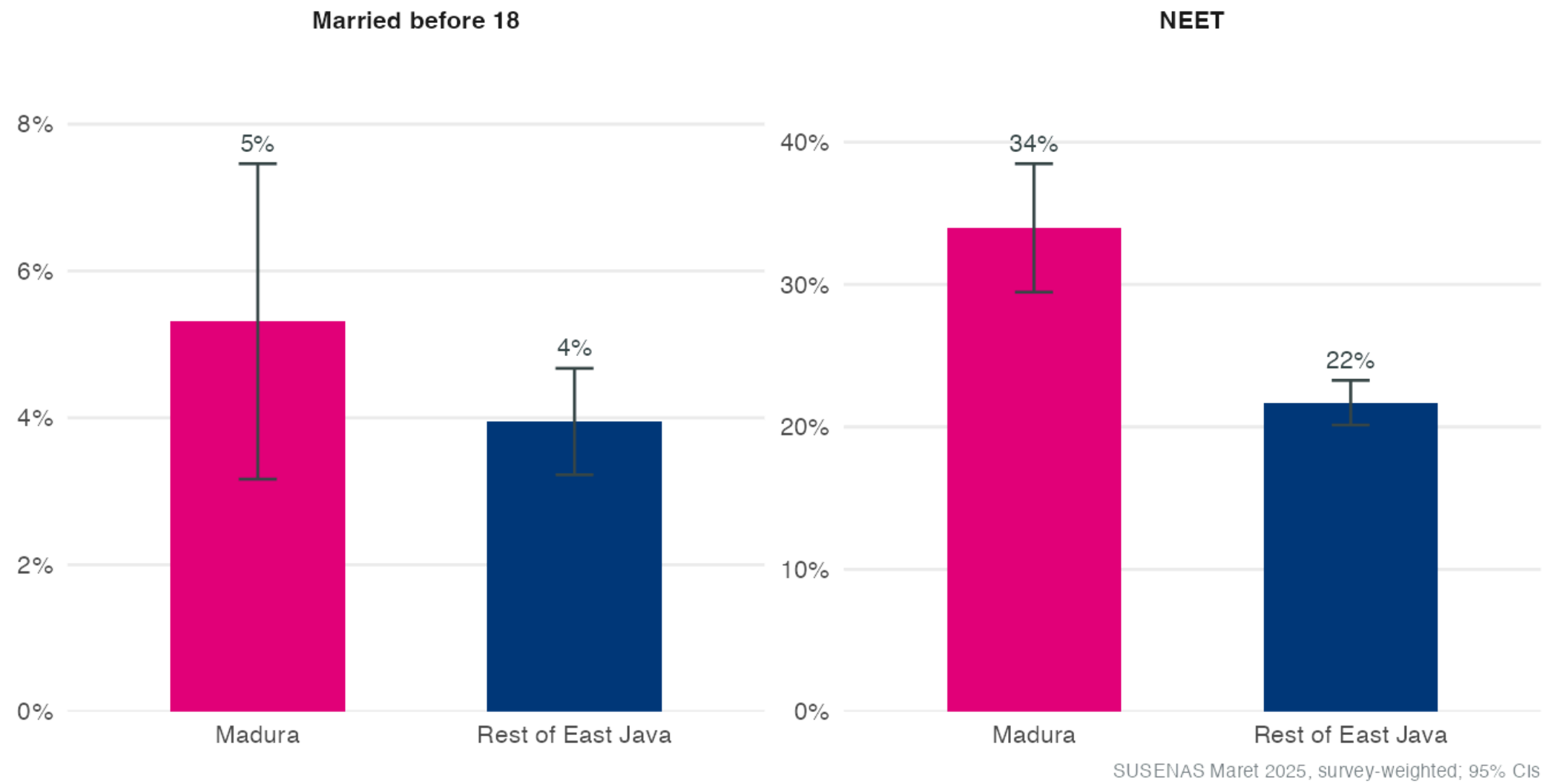


SUSENAS Maret 2025 — East Java (prov 35)

About 288,000 girls are NEET in excess of the boys' rate. Each would be expected to earn about USD 14,100 more over a working lifetime (present value) if she had the boys' NEET rate.

Madura has higher NEET for girls and boys, with a similar gender gap

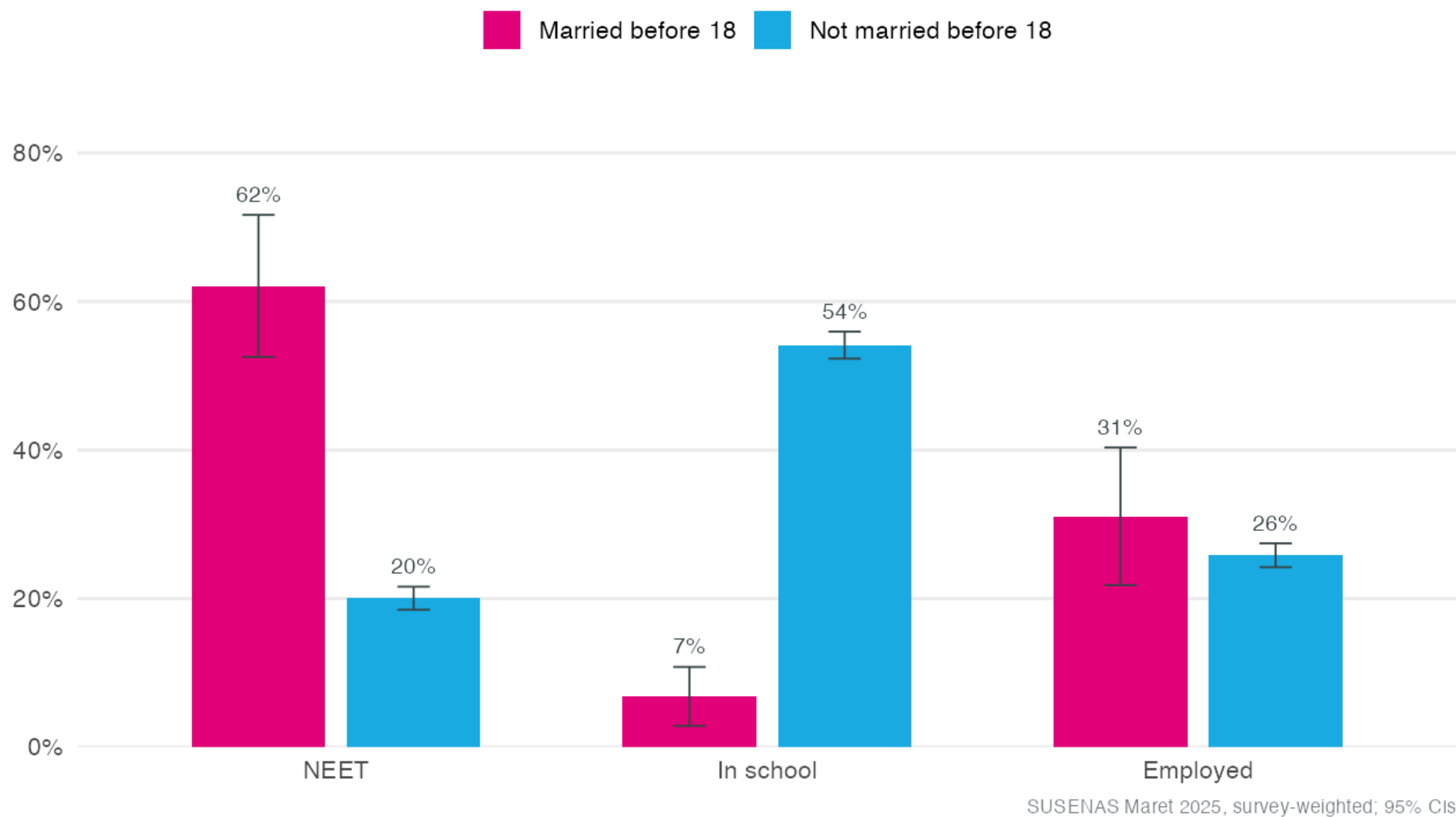
Girls 15-24: Madura (Bangkalan, Sampang, Pamekasan, Sumenep) vs rest of East Java



Girls' NEET is 34% in Madura (Bangkalan, Sampang, Pamekasan, Sumenep) against 22% in the rest of East Java; boys' NEET is 23% against 12%. The gender gap is similar (11 vs 10 points), and fewer girls are in school (45% vs 52%). Madura is a priority area for a pilot.

Child marriage and outcomes: rest of East Java

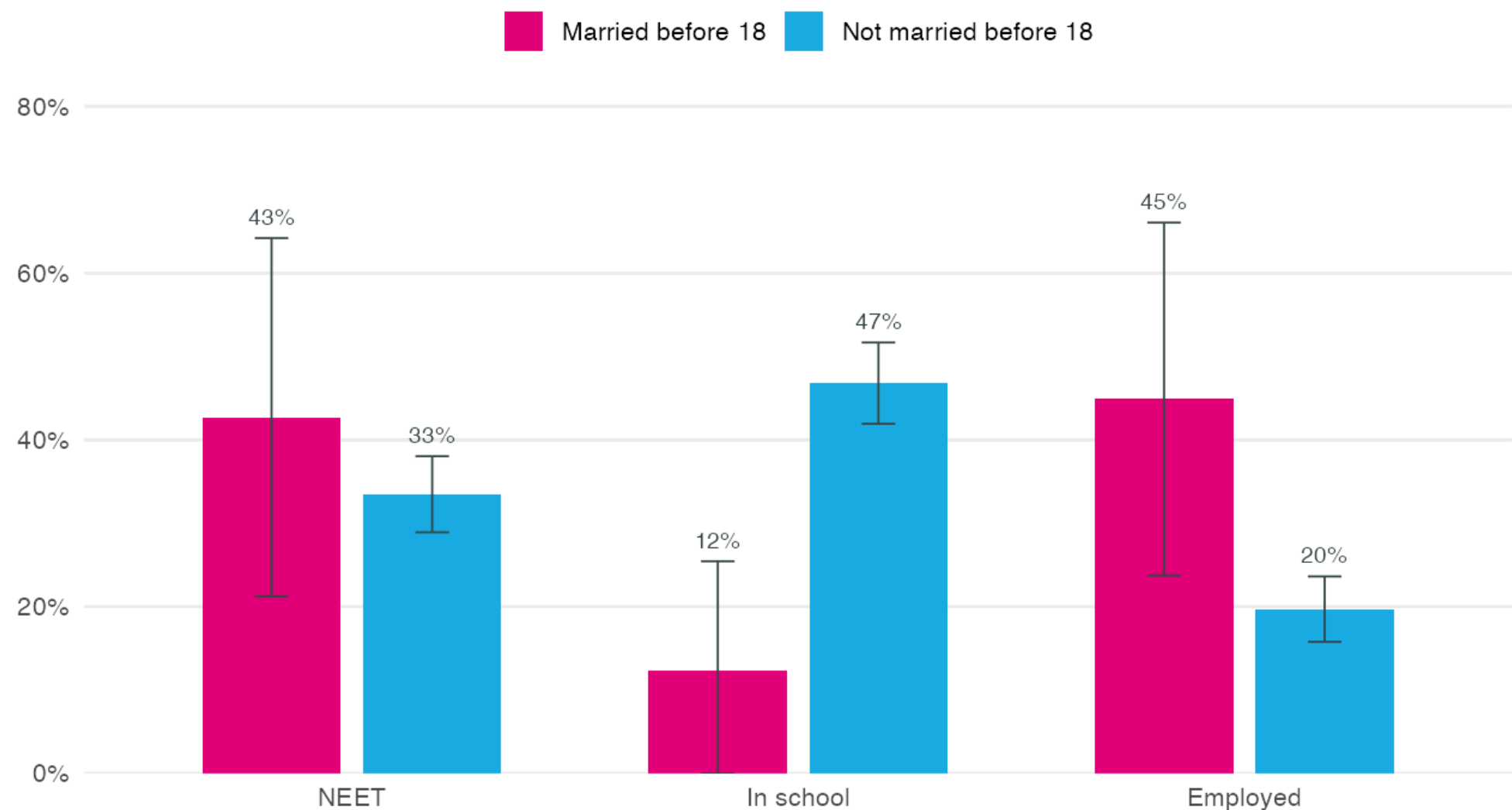
Girls 15-24, Rest of East Java



Girls 15–24 outside Madura, by whether they married before 18, survey-weighted (95% CIs). Regression tests by region are in the appendix. Early pregnancy before 18 overlaps almost entirely with child marriage in this data, so only child marriage is shown. Association, not causation.

Child marriage and outcomes: Madura

Girls 15-24, Madura

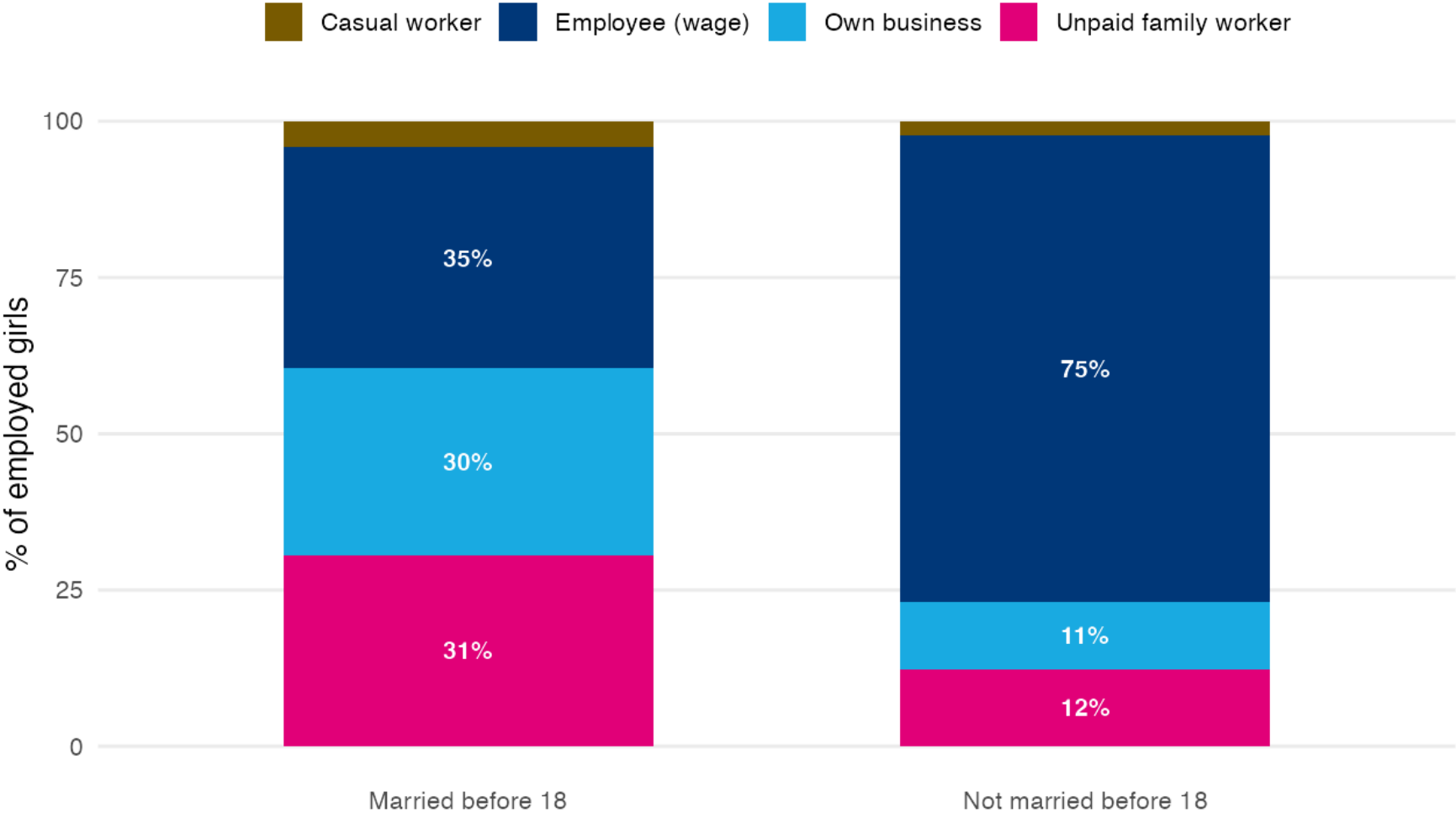


SUSENAS Maret 2025, survey-weighted; 95% CIs

Girls 15–24 in Bangkalan, Sampang, Pamekasan and Sumenep. Fewer girls, so wider intervals. Association, not causation.

Married girls who work are mostly in unpaid family work or own businesses

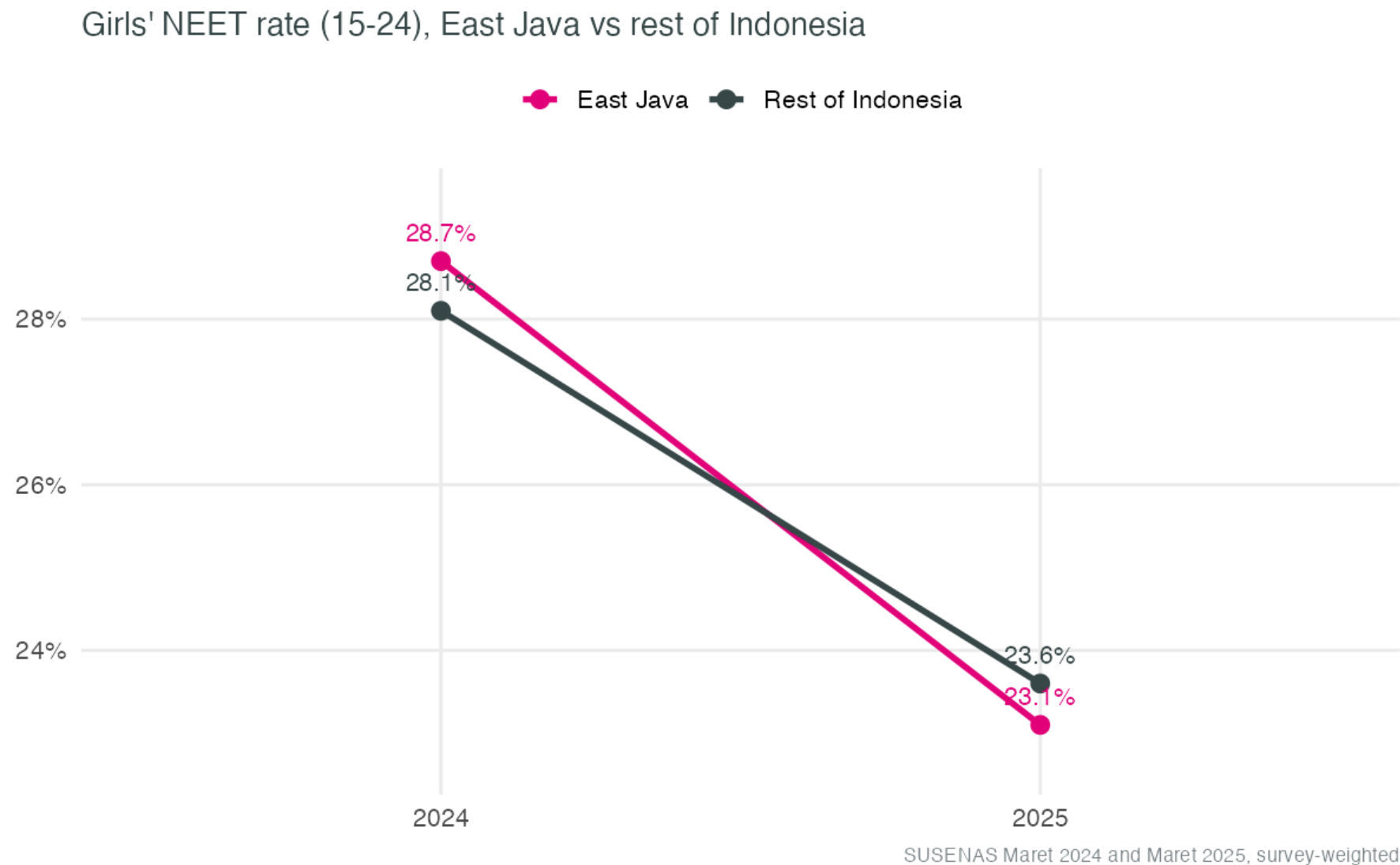
Type of work among employed girls 15-24, by child-marriage status



SUSENAS Maret 2025, East Java, survey-weighted

Among employed girls married before 18, 35% are wage employees (others 75%) and 31% are unpaid family workers (others 12%); 22% work under 20 hours a week (others 12%). Their employment is lower-quality work. Based on 78 employed married girls, so indicative. SUSENAS has no earnings; the appendix shows earnings by type of work for young women from IFLS5.

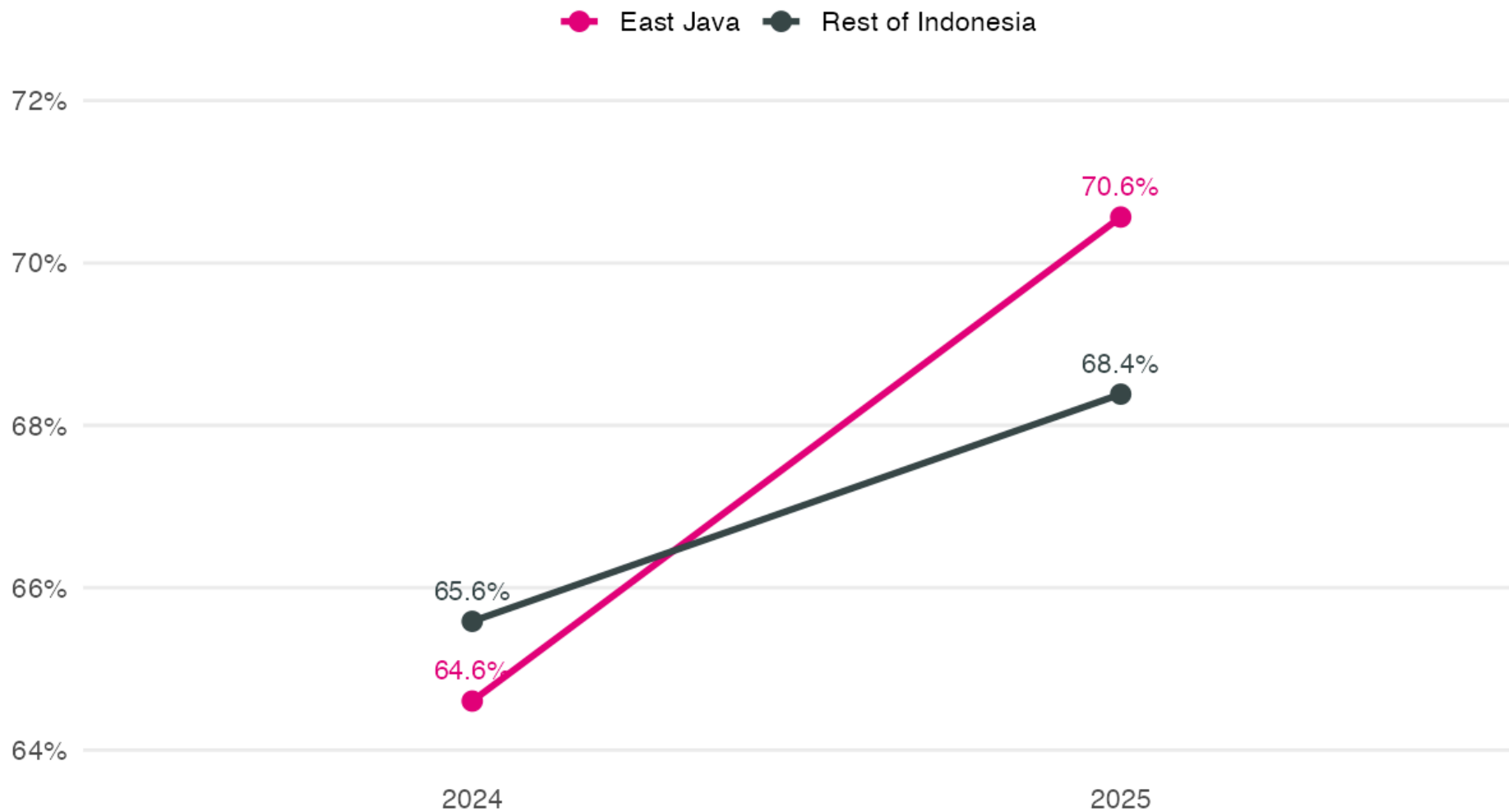
Girls' NEET fell from 2024 to 2025, slightly more than in the rest of Indonesia



Girls' NEET fell 5.6 points in East Java and 4.5 in the rest of Indonesia; child marriage fell 1.3 and 0.7 points. Tertiary enrolment at 18–20 rose 3.2 points, less than elsewhere (5.0). Most of the NEET decline is national, partly reflecting a change in the survey's activity question, so it cannot be attributed to a programme. A district-level comparison with programme districts is pending.

Senior-secondary enrolment also rose, in East Java and nationally

Girls 16-18 enrolled in senior secondary, East Java vs rest of Indonesia



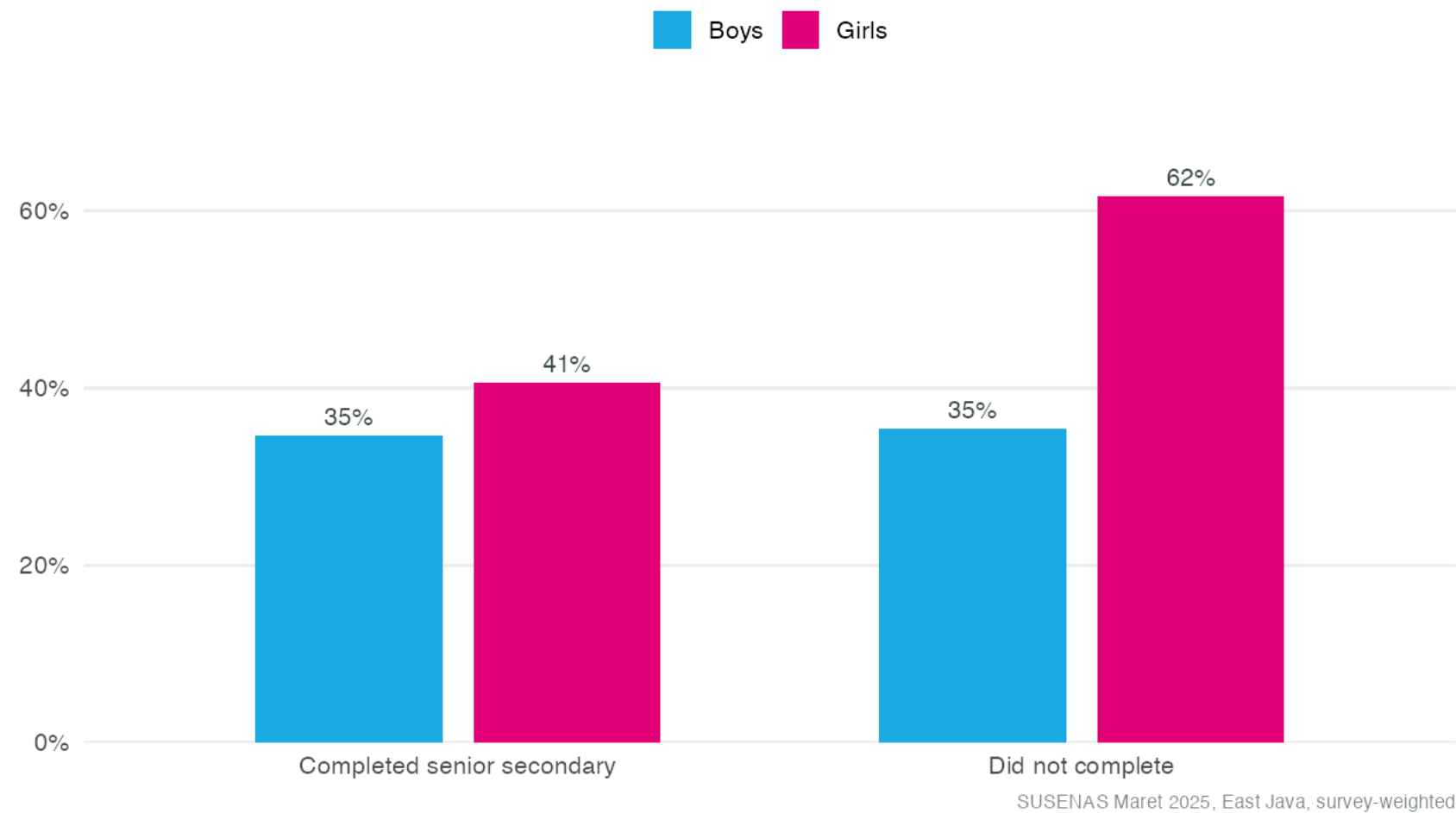
SUSENAS Maret 2024 and 2025, survey-weighted

Girls 16–18 enrolled in senior secondary, 2024 and 2025. A rise shared with the rest of Indonesia points to national drivers (such as PIP) rather than something specific to East Java.

2 · Theory of change

Completing senior secondary is associated with 20 points lower NEET for girls

NEET rate among 18-20 year olds who have left school, by senior-secondary completion



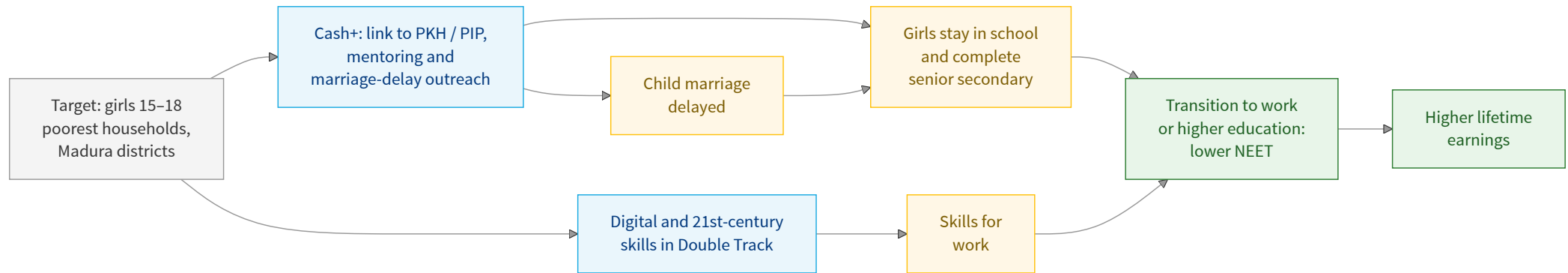
Among 18–20 year olds who have left school, girls who completed senior secondary have a NEET rate of 41%, against 62% for those who did not. Adjusted for age and area (girls 18–24), completion is associated with a 20-point lower NEET rate ($p<0.001$). For boys there is no difference. Keeping girls in school to completion is the programme’s main lever.

Evidence from Indonesia

Programme	Design	Result	Relevance
PKH cash transfer (Priebe & Sumarto, 2025)	Quasi-experimental	Girls’ junior-secondary enrolment +5.2pp; child marriage –3.5pp	A cash component reduces child marriage
PIP education assistance (Ulfa & Rezki, 2024)	Quasi-experimental	School attendance +11.4pp	Anchors the effect used in the ROI
PIP, girls’ senior secondary (Caniago, 2021)	Quasi-experimental	Girls’ senior-secondary enrolment +24.6pp	Targets the stage where girls drop out
Vocational training (Pritadrajati, 2022)	Quasi-experimental	Female LFPR +3.4pp; income +5.6%	Skills translate into work

In SUSENAS 2025, girls in PKH households have lower child-marriage rates than other girls (3% vs 4–6%), even though PKH targets poorer households. Consistent with the PKH evidence, though not a causal estimate.

Theory of change



Each arrow is backed by evidence from Indonesia: cash transfers delay girls' marriage and raise their enrolment (PKH, girls); education assistance raises girls' senior-secondary enrolment (PIP, girls); completion is associated with 20 points lower NEET for girls (this analysis); skills raise women's employment and income (vocational training, women). Mentoring and outreach have only international evidence for girls (ELA, Room to Read). The cash component links girls to existing PKH and PIP rather than creating a new transfer.

3 · Return on investment

Key assumptions

Assumption	Value	Source
Reduction in girls' NEET	5 points	PIP attendance effect (+11.4pp), 45% assumed to come from girls who would be NEET
Extra schooling for a girl kept in school	3 years	Senior-secondary cycle
Earnings gain per year of schooling	8.0%	IFLS5 wage return for women, controlling for region, parents' education and selection into paid work
Female earnings	Rp20 million/year	Below East Java minimum wage 2025, reflecting informal work
Female labour force participation	52%	Sakernas, East Java
Earnings growth / discount rate	2% / 3% per year	Career growth; World Bank education default
Skills earnings premium, all exposed girls who work	5.6% for 10 years	Vocational training raised women's income by 5.6% (Pritadrajati 2022)

How the return is built

Step	Value
Extra years of schooling for a girl kept in school	3
Earnings gain per year of schooling	8.0%
Earnings gain for that girl	24%
Female earnings (USD/yr) x labour force participation	1,266 x 52%
Extra earnings per year (USD)	157
Present value over the working life (USD per girl moved out of NEET)	4,035
Girls exposed per girl moved out of NEET (1 / NEET reduction)	20
Cost per girl moved out of NEET (USD)	4,002
Return per USD 1, retention channel only	1.01
Skills premium for all exposed girls who work (USD per exposed girl, PV)	304
Return per USD 1, both channels	2.53

Every USD 1 invested returns about USD 2.53

USD 2.53

return per USD 1, both channels

USD 1.01

return per USD 1, retention
channel only (floor)

USD 200

cost per girl (3 years)

**USD
4,002**

cost per girl moved out of NEET

- **Retention channel:** 5 in 100 exposed girls stay in school and avoid NEET; each gains about three years of schooling and a lifetime of higher earnings.
- **Skills channel:** every exposed girl who works earns more, as vocational training has shown in Indonesia.
- Both channels value earnings only. Delayed marriage, health, intergenerational and fiscal benefits are excluded, so this is a lower bound.

The return depends mainly on the programme’s effect on NEET

Scenario	Retention benefit (USD/girl)	Skills premium benefit (USD/girl)	Total benefit (USD/girl)	Cost (USD/girl)	BCR	Cost per NEET case averted (USD)
Retention channel only (floor)	202	0	202	200	1.01	4,002
Central	202	304	506	200	2.53	4,002
Central, earnings at the 2025 minimum wage	280	422	702	200	3.51	4,002
Central, with Cash+ stipend for the poorest 40%	202	304	506	359	1.41	7,176

The floor is the retention channel alone; the central case adds the skills premium; the other rows show what happens at the 2025 minimum wage, and if a Cash+ stipend for the poorest 40% is added to the cost. With other assumptions varied one at a time, the central return stays between USD 1.90 and USD 3.26 (appendix).

4 · Key messages

Key messages

1. Girls in East Java stay in school as long as boys but are close to twice as likely to be NEET, a wider gap than nationally. The gap costs about USD 4.1 billion in forgone lifetime productivity.
2. Madura is the priority area: NEET is higher for both sexes, and fewer girls are in school.
3. Child marriage is strongly associated with leaving school and with lower-quality work; poorer girls face higher risk.
4. Completing senior secondary is associated with 20 points lower NEET for girls, which makes school completion the programme's main lever.
5. A package of skills, mentoring and links to PKH/PIP costing about USD 200 per girl returns about USD 2.53 per USD 1 on earnings alone, and at least USD 1.01 even if its only effect is keeping girls in school.

Appendix

Regression results: child marriage and outcomes

Hypothesis	Effect (pp)	SE	p	N
Child marriage → NEET	28.7	4.6	0.0000	6060
Child marriage → in school	-27.5	3.0	0.0000	6060
Child marriage → employment	-1.2	4.6	0.7942	6060

Linear-probability models, girls 15–24, survey-weighted, PSU-clustered. **Controls: age and urban/rural.** Poverty is not controlled for because the 2025 expenditure module is not yet available. The association with employment is small and not statistically significant. Association, not causation.

Child marriage and outcomes by region: regression results

Region	Outcome	Effect (pp)	SE	p	N
Madura	Employed	18.7	11.1	0.093	664
Madura	NEET	3.1	12.3	0.801	664
Madura	In school	-21.8	9.8	0.026	664
Rest of East Java	Employed	-4.6	5.1	0.361	5396
Rest of East Java	NEET	33.2	5.0	0.000	5396
Rest of East Java	In school	-28.5	3.1	0.000	5396
All East Java	Employed	-1.2	4.6	0.794	6060
All East Java	NEET	28.7	4.6	0.000	6060
All East Java	In school	-27.5	3.0	0.000	6060

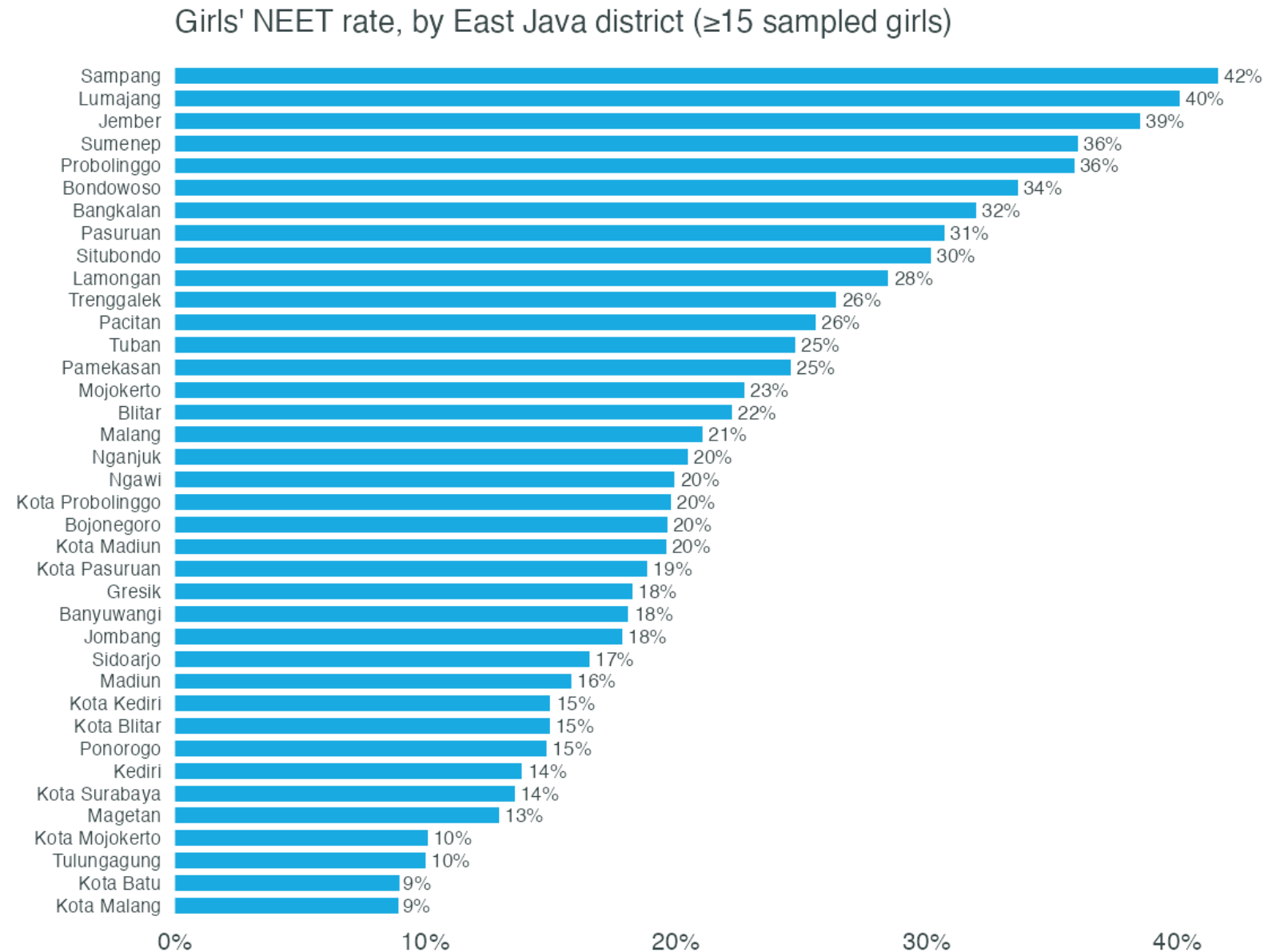
Linear-probability models, girls 15–24, survey-weighted, PSU-clustered; controls for age and urban/rural. Effects in percentage points.

Earnings by type of work, young women (IFLS5)

status	n	Share with earnings reported	Median monthly earnings (IDR)
5:Private worker	1440	95	1200000
6:Unpaid family worker	614	0	NA
1:Self-employed	355	0	NA
2:Self employed with unpaid family/temporary worker	315	0	NA
4:Government worker	222	93	887500
8:Casual worker not in agriculture	146	85	465000
7:Casual worker in agriculture	57	82	200000
3:Self-employed with employees/permanent workers	19	0	NA

Women 15–30, IFLS5 (2014/15 prices). SUSENAS collects no earnings. Unpaid family workers have no earnings by definition.

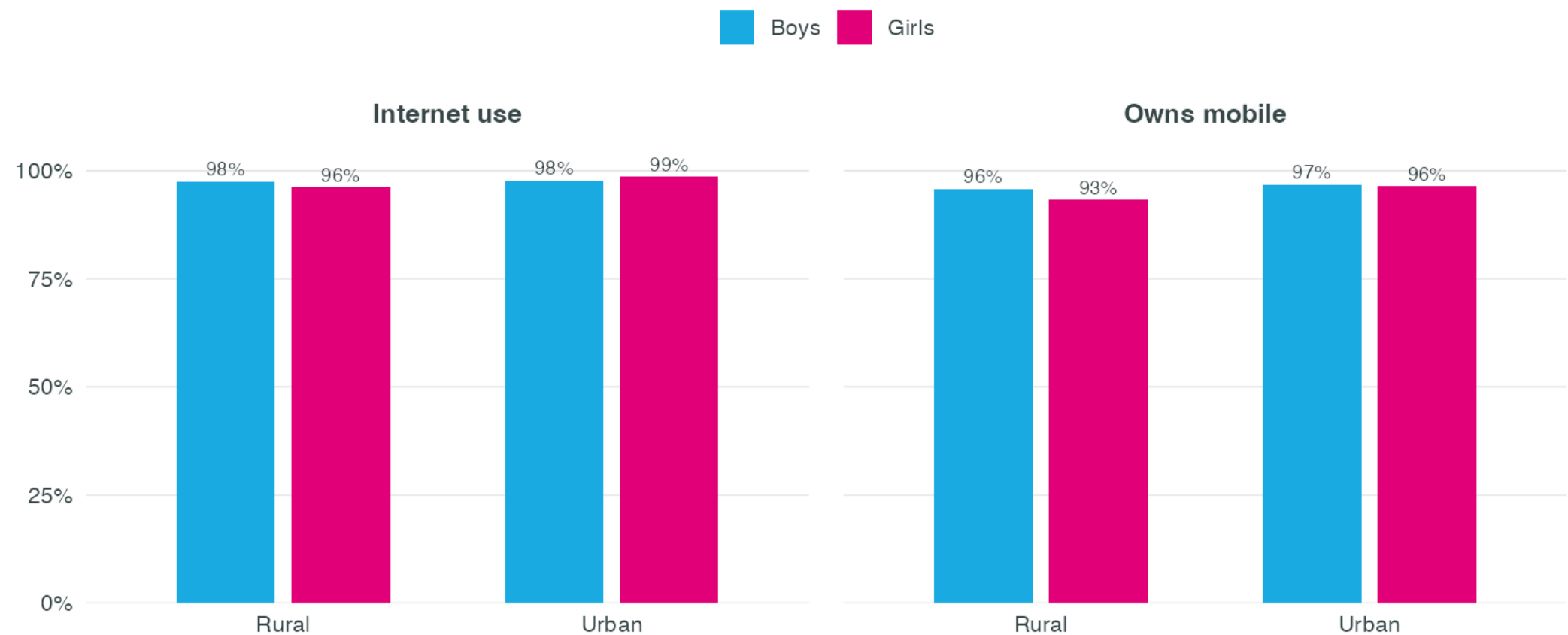
NEET among girls, by district



Source: SUSENAS Maret 2025 — East Java (prov 35). Weighted, adolescents 15–24.

Digital access and use

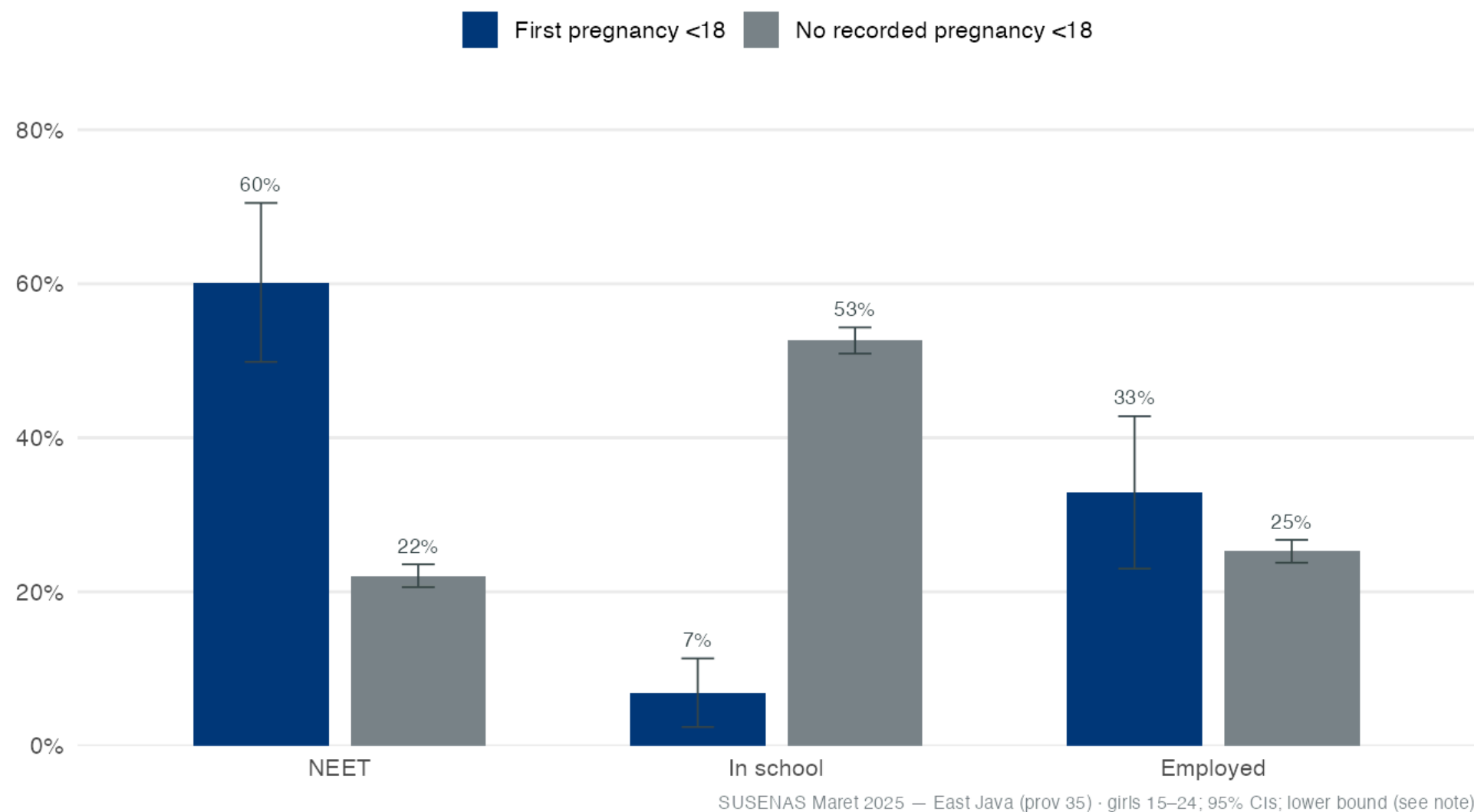
Digital access and use, by sex and area



Source: SUSENAS Maret 2025 — East Java (prov 35). Weighted, adolescents 15–24.

Early pregnancy and outcomes

Girls with an early pregnancy are far likelier to be NEET and out of school



Recorded only for ever-married women, so a lower bound. Girls with a first pregnancy before 18 are almost all girls married before 18, so the main deck reports child marriage only.

Child marriage and NEET by household PKH receipt

region	PKH	n	Married_u18	NEET
Madura	Not PKH	509	6.1	35.9
Madura	PKH household	155	3.0	28.5
Rest of East Java	Not PKH	4661	4.1	22.1
Rest of East Java	PKH household	735	3.2	19.5

Girls 15–24, SUSENAS 2025. PKH targets poorer households, so lower rates among recipients are consistent with a protective effect, but are not a causal estimate.

Cost components

Component	USD/girl/yr	Source
Digital content & platform	10	Kemendikbudristek Merdeka Mengajar platform; INOVASI digital-learning unit costs
Teacher training (annualised)	15	Double Track PD days, annualised
Mentoring & socio-emotional	10	ELA/Skills4Girls facilitator costs
School connectivity/devices (shared)	15	Shared labs, no 1:1 devices
Ancillary: marriage-delay + flexible learning	8	ToR guidance: outreach to delay marriage, flexible options for caregivers
Annual direct cost per girl	58	Sum
Total per girl (3 years + 15% overhead)	200	Programme cost per girl

These are benchmark unit costs from comparable programmes, not quotes from Double Track; they are placeholders until the provincial government shares actual budgets. The transfer itself is not costed because the Cash+ component links girls to existing PKH and PIP entitlements; a new stipend for the poorest 40% at the PIP senior-secondary rate would add about USD 46 per girl per year (ROI table).

Sensitivity of the return to each assumption

Parameter	BCR at low value	BCR at high value
Years the skills premium lasts	1.79	3.91
Skills earnings premium (all exposed girls who work)	1.01	2.53
Annual cost per girl (sensitivity)	3.26	1.96
Female annual wage (East Java)	1.90	3.16
NEET reduction per exposed girl	2.03	3.03
Discount rate	1.96	2.91
Mincer return per year of edu	2.23	3.07
Female labour force participation	2.19	2.92
Real earnings growth	2.15	2.77

Each assumption varied on its own, others held at central values.

What the return counts

Counted

Higher schooling leads to a lifetime earnings premium for girls moved out of NEET, using the female return to schooling over a discounted working life.

Excluded (so the return is a lower bound)

Delayed child marriage · health and productivity spillovers · intergenerational gains · fiscal returns · broader empowerment

Comparable international evidence

Programme	Design	Girls / age	Focus	Key result	Fit for East Java
Adolescent girls' wellbeing, Bergström & Özler 2021	Review (104 studies)	Adolescent girls	Multi-sector interventions	Integrated packages beat single-sector ones	Synthesis backbone for the ToC
NEET in Indonesia, Faizah, Isbah & Azca 2025	Qualitative	Youth (gendered)	Drivers of girls' NEET	Non-education factors (norms, care, marriage) drive girls' NEET	Indonesia-specific NEET context
ELA girls' clubs, Bandiera 2018 (Sierra Leone)	RCT	Girls 12–25	Vocational + life skills	Enrolment +8.5pp; more income-generating activity	Closest Skills4Girls-in-Double-Track model
Female digital-employment bootcamp, Alzúa 2020 (Argentina/Colombia)	RCT	Women 18–50	Coding for employment	Large coding-skill gains	Core digital-skills → jobs arm
Irumi, Näslund-Hadley 2025 (Paraguay)	RCT	Pupils	Computational thinking	+0.09 SD CT	Structured CT pedagogy
Robotics & Coding, Gülen 2025 (Turkey)	RCT	Girls 14–16	Robotics / coding	Higher STEM-career orientation	Girls' STEM, secondary age
Samata, Prakash 2019 (India)	RCT	Poor girls 13–16	Secondary-transition support	More girls transition; norm shift	South Asia; senior-secondary transition
Room to Read GEP, Edmonds 2021 (India)	RCT	Girls	Life skills + mentoring	Dropout –3.5 to –4.1pp	Mentoring/pedagogy layer cuts dropout
Negotiating a Better Future, Ashraf 2025 (Zambia)	RCT	Girls ~14	Negotiation skills	+0.26 years schooling (10-yr)	Durable, cheap soft-skills layer
OLPC / Plan Ceibal, de Melo 2014 (Uruguay)	Quasi / review	Pupils	Laptops (hardware-led)	Learning ≈ null (–0.05 SD)	Caution: hardware alone fails → fund pedagogy

Returns to schooling: specification and results

Wage-based (IFLS5 2014/15, adults 20–60, estimated separately for women and men, weighted):

$$\ln(w_i) = \beta_0 + \beta_1 S_i + \beta_2 \text{Exp}_i + \beta_3 \text{Exp}_i^2 + \beta_4 \text{Urban}_i + \gamma_p + \phi_f + \phi_m + \delta \lambda_i + \varepsilon_i$$

w : monthly wage; S : years of schooling; Exp : potential experience (age minus schooling minus 6); γ_p : province fixed effects; ϕ_f, ϕ_m : father’s and mother’s education (fixed effects); λ : selection term from a first-stage model of working, in which children under 5 in the household predict whether a woman works but not her wage (Heckman two-step).

Sex	Model	Return per year	SE	N
Women	(1) Baseline	12.0%	0.5	3830
Women	(2) + Region	12.2%	1.1	3830
Women	(3) + Family background	10.9%	1.1	3828
Women	(4) + Selection (Heckman)	8.0%	1.2	3828
Men	(1) Baseline	9.4%	0.3	6844
Men	(2) + Region	8.7%	0.6	6843
Men	(3) + Family background	8.0%	0.5	6842
Men	(4) + Selection (Heckman)	10.1%	1.0	6842

The ROI uses the women’s estimate with all controls (model 4, 8.0%). The consumption-based return from SUSENAS 2024 (5.6% for women; household per-capita expenditure regressed on schooling, schooling², age, age² and urban) is the low case in the sensitivity analysis.

Data sources

- **SUSENAS Maret 2025** (BPS): NEET, school attendance and completion, child marriage, early pregnancy, type of work and hours, PKH receipt and digital access, for East Java and all provinces. East Java: 98,738 individuals, 12,599 aged 15–24, of whom 6,060 girls.
- **SUSENAS Maret 2024** (BPS): the 2024 comparison, and per-capita expenditure for the consumption-based return to schooling.
- **IFLS5 2014/15**: wages for the wage-based return to schooling.
- **Sakernas** (BPS): female labour force participation in East Java.
- **UMP Jawa Timur 2025**: benchmark for female earnings.
- **Published evaluations** of PKH, PIP and vocational training: the programme effect used in the return on investment.

All SUSENAS estimates are survey-weighted, with PSU/strata-clustered standard errors.

Definitions

- **NEET (15–24)**: not attending school ([R611](#)) and not working ([R703_A](#), [R704](#)).
- **Married before 18**: age at first marriage ([R409](#)) under 18.
- **Senior-secondary completion**: highest certificate ([R615](#)) at SMA/SMK/MA level or above.
- **Type of work**: status in main job ([R706](#)); hours ([R707](#)).
- **Madura**: Bangkalan, Sampang, Pamekasan, Sumenep.

The 2025 questionnaire replaced the single main-activity question with a multi-select block, so year-on-year changes partly reflect the new question; comparisons with the rest of Indonesia use the same definition in both places.